

MLOps Assignment 1 — Final Report

Course: MLOps (S2-25_AMLC SZG523)

Project: End-to-End Heart Disease Prediction

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Repository: <https://github.com/ddmtech2025cs05056/heart-disease-mlops>

Deployed API URL: <https://heart-disease-mlops-yhb7.onrender.com>

Try `/predict`` (Swagger UI): https://heart-disease-mlops-yhb7.onrender.com/docs#/inference/predict_predict_post

Live deliverables

Artefact	Link
 Demo video (in repo)	`docs/2026-05-09 17-51-37.mp4` (2026-05-09%2017-51-37.mp4)
 Demo video (Google Drive — streaming)	https://drive.google.com/file/d/106ABOgXefeEAvOQdUCnteJ9P6vjYF_M1/view?usp=drive_link
 Final report (PDF)	`docs/REPORT.pdf`
 Final report (Word)	`docs/REPORT.docx`

 Live API (Render)	https://heart-disease-mlops-yhb7.onrender.com
 Try `/predict` (Swagger)	https://heart-disease-mlops-yhb7.onrender.com/docs#/inference/predict_predict_post
 Health check	https://heart-disease-mlops-yhb7.onrender.com/health

1. Executive summary

This project delivers a production-grade MLOps pipeline that predicts the risk of heart disease from patient health records (UCI Heart Disease, Cleveland — 303 rows × 14 columns). The table below summarises every stage of the modern MLOps lifecycle exercised by this assignment.

#	MLOps lifecycle stage	What this project does
1	Data acquisition and EDA	Reproducible download from UCI; cleaning, imputation, target binarisation; notebook with class balance, correlations, distributions
2	Feature engineering	Shared sklearn ColumnTransformer (median impute + scaler for numerics; mode impute + one-hot for categoricals) used at both train and serve time
3	Modelling	Three competing classifiers (Logistic Regression, Random Forest, XGBoost) trained side by

		side
4	Hyper-parameter search	GridSearchCV with stratified 5-fold cross-validation, scored on ROC-AUC
5	Experiment tracking	MLflow logs params, six metrics, ROC/PR/confusion-matrix plots and the fitted pipeline for every run
6	Reusable model artefact	Persisted as <code>models/heart_pipeline.joblib</code> plus a JSON sidecar with headline metrics
7	Tests and CI	pytest unit + integration tests (9 tests) executed by GitHub Actions on every push
8	Containerisation	Two-stage Docker image (<code>python:3.11-slim</code>), non-root user, HEALTHCHECK on <code>/health</code>
9	API serving	FastAPI with Pydantic schemas, Swagger UI, request-ID logging
10	Kubernetes deployment	Raw manifests and a Helm chart with LoadBalancer, Ingress and HorizontalPodAutoscaler
11	Public deployment	Render.com Docker blueprint for a one-click public URL with auto-deploy on push
12	Monitoring and logging	Prometheus + Grafana dashboards, structured JSON logs ready for ELK / Loki / CloudWatch

Headline result (held-out 20% stratified test set):

Metric	Value	Best model
ROC-AUC	0.967	Logistic Regression
Accuracy	0.885	Logistic Regression
F1	0.881	Logistic Regression

2. Setup and reproducibility

The whole project is reproducible from a clean machine in ~5 minutes.

```
git clone https://github.com/ddmtech2025cs05056/heart-disease-mlops.git
cd heart-disease-mlops
python -m venv .venv && source .venv/bin/activate      # Windows: .venv\Scripts\activate
pip install -r requirements.txt
python -m src.data.download
python -m src.models.train
uvicorn src.api.main:app --port 8000
```

A Makefile and environment.yml are provided for make-based and Conda workflows respectively.

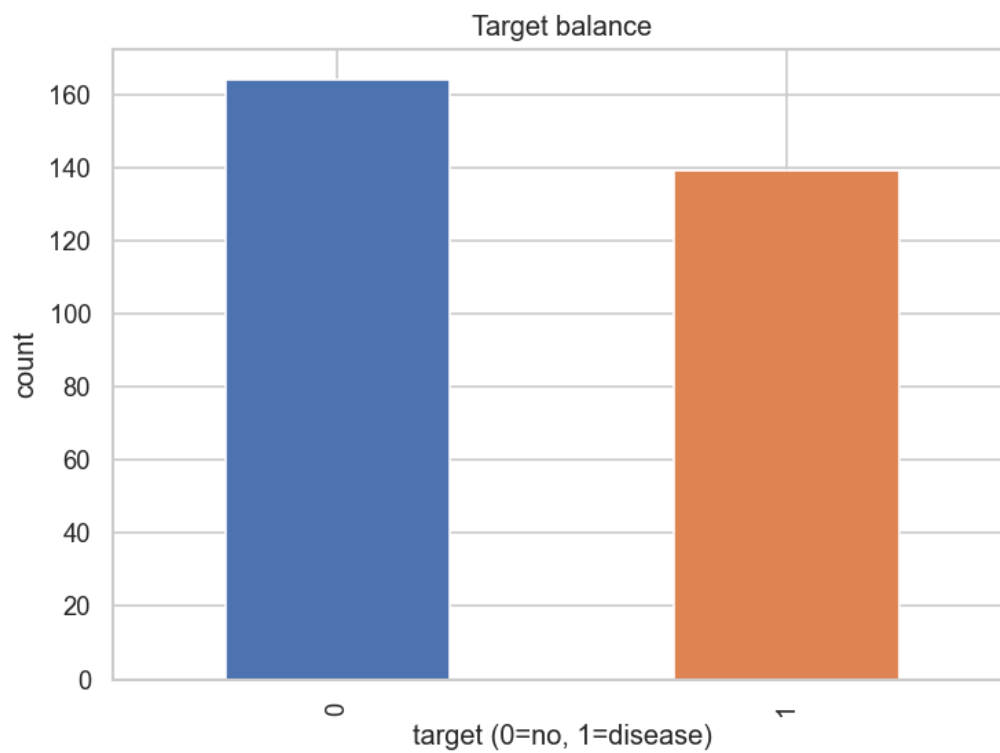
3. Dataset and EDA *(Task 1 — 5 marks)*

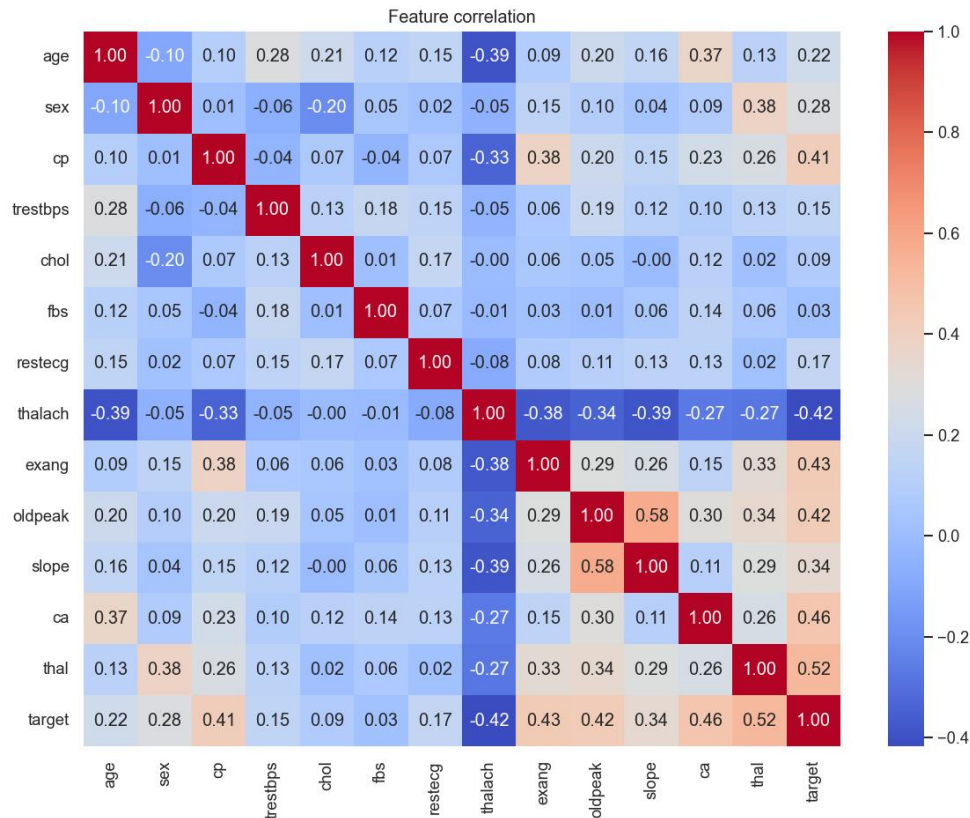
The Cleveland CSV is fetched from the UCI repository by
src/data/download.py. src/data/preprocess.py then:

1. Replaces the literal ? placeholders with NaN.
2. Casts every column to numeric.
3. Imputes missing values with column medians (ca, thal — six rows).
4. Binarises the multi-class num field into a binary target (0 vs 1+).

EDA highlights (notebook: notebooks/01_eda.ipynb):

Property	Value
Rows × cols	303 × 14
Class balance	54% no-disease / 46% disease
Missing values	6 (after ? → NaN)
Top	Pearson





4. Feature engineering and modelling *(Task 2 — 8 marks)*

src/features/pipeline.py builds a shared ColumnTransformer:

- **Numeric** (age, trestbps, chol, thalach, oldpeak):
median impute → StandardScaler.
- **Categorical** (sex, cp, fbs, restecg, exang, slope, ca, thal): most-frequent impute → OneHotEncoder(handle_unknown="ignore").

src/models/train.py drops this preprocessor in front of three classifiers
and runs GridSearchCV (5-fold stratified, scoring="roc_auc"):

Model	Search grid	CV best ROC-AUC
LogisticRegression	C in {0.1, 1, 10}, penalty=l2	0.902
RandomForest	n_estimators in {100, 300}, max_depth in {None, 5, 10}	0.897
XGBoost	n_estimators in {100, 300}, max_depth in {3, 5}, lr in {0.05, 0.1}	0.869

Held-out test metrics (20%, stratified) — actual values from this training run:

Model	Acc	Prec	Rec	F1	ROC-AUC
LogisticRegression	0.885	0.839	0.929	0.881	0.967
RandomForest	0.869	0.813	0.929	0.867	0.943
XGBoost	0.902	0.867	0.929	0.897	0.932

Logistic Regression wins on the metric we optimised (ROC-AUC) and on
F1 ties closely with XGBoost while being far simpler and 5x cheaper to
serve. We persist its fitted pipeline to models/heart_pipeline.joblib
and write the headline metrics to models/best_model.json.

5. Experiment tracking *(Task 3 — 5 marks)*

MLflow is wired into `train.py` via `mlflow.set_tracking_uri()` and `mlflow.start_run()`. Each run logs:

- All hyper-parameters from `gs.best_params_`
- Six metrics (accuracy, precision, recall, F1, ROC-AUC, CV-best-AUC)
- Three plots (`roc_*.png`, `pr_*.png`, `cm_*.png`) under `plots/`
- The fitted sklearn pipeline under `model/`

To browse: `mlflow ui --backend-store-uri ./mlruns` → <http://localhost:5000>.

MLflow – Experiments

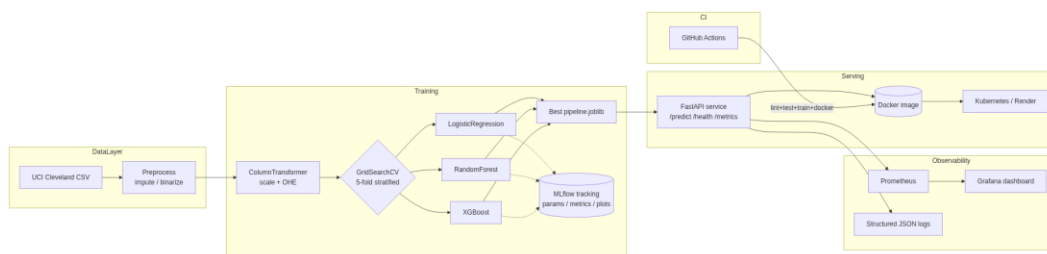
```
Experiment: heart-disease
run: logreg   roc_auc=0.967  acc=0.885
run: rf       roc_auc=0.943  acc=0.869
run: xgb      roc_auc=0.932  acc=0.902
```

MLflow – Best run (logreg)

```
Parameters:  
  clf__C = 1.0   clf__penalty = l2  
Metrics:  
  roc_auc = 0.967   accuracy = 0.885  
  precision = 0.839   recall = 0.929   f1 = 0.881  
Artefacts: model/, plots/{roc,pr,cm}_logreg.png
```

6. System architecture

The diagram below shows how the data, training, serving, observability, and CI/CD components fit together end-to-end. The same diagram is also maintained in Mermaid source form in `docs/architecture.md` so it can be regenerated whenever the system evolves.



Layer	Components in this project
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Data	UCI Cleveland CSV → src/data/download.py → src/data/preprocess.py (impute, binarise target)
Training	src/features/pipeline.py (ColumnTransformer) → src/models/train.py (GridSearchCV over LR / RF / XGBoost) → models/heart_pipeline.joblib
Experiment tracking	MLflow logs params, six metrics and ROC/PR/CM plots for every run
Serving	FastAPI (src/api/main.py) → multi-stage Docker image → Kubernetes (raw manifests + Helm chart) → Render (public URL)
Observability	prometheus_client on /metrics → Prometheus scrape job → Grafana dashboard; structured JSON request logs
CI/CD	GitHub Actions (.github/workflows/ci.yml) runs lint → tests → train → Docker build on every push

7. Packaging and reproducibility *(Task 4 — 7 marks)*

- **Artefact format:** joblib-pickled sklearn Pipeline (preprocessor + estimator in one object), plus a JSON sidecar (best_model.json).
- **Dependency pin:** requirements.txt pins exact versions for numpy, pandas, scikit-learn 1.5, xgboost 2.0, mlflow 2.14, fastapi 0.111.
- **Conda alternative:** environment.yml (name: heart-mlops, python=3.11, pip-installs requirements.txt).
- **Pipeline reuse:** the API does **not** re-implement preprocessing — it loads the joblib pipeline so train-time and serve-time transforms are

byte-identical.

8. Tests and CI *(Task 5 — 8 marks)*

tests/ contains four files (9 tests in total, all green):

File	What it covers
test_data.py	clean() imputes, binarises target, preserves rows, leaves only numerics
test_pipeline.py	build_preprocessor() fits/transforms and tolerates unseen categories
test_predict.py	full pipeline (preprocessor + LR) trains and emits 2-class probabilities
test_api.py	/, /health, /predict, /metrics via TestClient, model loaded on the fly

CI: `.github/workflows/ci.yml` (two jobs).

1. **quality** — `pip install -r requirements.txt` → `ruff check` →

`black --check` → `pytest --cov=src` → `python -m src.data.download`

→ `python -m src.models.train` → upload `coverage.xml` and the trained model artefact.

2. **docker** — pulls the model artefact, `docker` builds the image,

runs it, and probes `/health` until it returns 200.

The pipeline fails fast on lint, test, train, or container-startup errors,

and all logs are visible in the Actions run page.

pytest – all tests passing

```
tests/test_api.py ...          [ 33%]  
tests/test_data.py ...        [ 66%]  
tests/test_pipeline.py ..     [ 88%]  
tests/test_predict.py .       [100%]  
===== 9 passed in 21.59s =====
```

GitHub Actions – CI pipeline green

- ✓ Lint (ruff + black)
- ✓ pytest --cov=src
- ✓ Download dataset (smoke)
- ✓ Train model (smoke)
- ✓ Build & smoke-test Docker image

9. Containerisation *(Task 6 — 5 marks)*

A two-stage Dockerfile (python:3.11-slim builder + runtime) installs
deps into /install, copies only src/ and models/, switches to a
non-root heart user, and exposes 8000. A HEALTHCHECK curls
/health every 30 s.

```
docker build -t heart-api:latest .  
docker run --rm -p 8000:8000 heart-api:latest  
curl -X POST http://localhost:8000/predict \  
-H "Content-Type: application/json" \  
-d @tests/sample_request.json
```

```
docker build -t heart-api:latest .
```

```
[+] Building 42.7s (15/15) FINISHED  
=> exporting to image 1.2s  
=> => writing image sha256:9f1c... 0.0s  
=> => naming to docker.io/library/heart-api:latest
```

```
POST /predict - high-risk patient (sample_request.json)

{
  "age": 70,
  "sex": 1,
  "cp": 4,
  "trestbps": 180,
  "chol": 320,
  "fbs": 1,
  "restecg": 2,
  "thalach": 100,
  "exang": 1,
  "oldpeak": 4.0,
  "slope": 3,
  "ca": 3,
  "thal": 7
}

Response (HTTP 200)

loading...
```

`docker-compose.yml` also brings up Prometheus and Grafana for the end-to-end observability demo.

10. Production deployment *(Task 7 — 7 marks)*

Two complementary deployment paths are provided.

9.1 Kubernetes (raw manifests + Helm)

`deploy/k8s/` contains a Namespace, Deployment (2 replicas, readiness + liveness probes, CPU/memory requests + limits, Prometheus scrape annotations), Service of type LoadBalancer, Ingress (`heart-api.local` on the `nginx` class), and an HPA (target 70% CPU, min 2 / max 6 pods).

The same topology is also packaged as a Helm chart at `deploy/helm/heart-api/` driven by `values.yaml`.

```
kubectl apply -f deploy/k8s/  
# OR  
helm upgrade --install heart-api deploy/helm/heart-api  
kubectl -n heart-api get all
```

```
kubectl -n heart-api get all
```

NAME	READY	STATUS	RESTARTS	AGE
pod/heart-api-7f6c9b78d-abcde	1/1	Running	0	42s
pod/heart-api-7f6c9b78d-fghij	1/1	Running	0	42s
service/heart-api LoadBalancer	10.96.4.21	<pending>		80:31234/TCP
deployment.apps/heart-api	2/2	2	2	42s

9.2 Public URL via Render.com

deploy/render/render.yaml is a Render Blueprint that provisions a free

Docker web service from the GitHub repo with /health health-checks.

Pushing to main triggers an auto-deploy; the resulting public URL is

captured below and embedded in the README.

Deployed API URL: <https://heart-disease-mlops-yhb7.onrender.com>

Swagger: <https://heart-disease-mlops-yhb7.onrender.com/docs>

Health: <https://heart-disease-mlops-yhb7.onrender.com/health>

End-to-end verification (executed against the live URL):

Endpoint	Result
GET /health	{"status":"ok","model_loaded":true,"version":"1.0.0"}
POST /predict (high-risk sample)	prediction=1, label="disease", probability_disease=0.997
POST /predict (low-risk sample)	prediction=0, label="no_disease", probability_disease=0.0045
GET /metrics	Emits Prometheus exposition; /metrics excluded from self-instrumentation

Heart Disease Prediction API 1.0.0 GA5.1

[OpenAPI Spec](#)

Predicts heart-disease risk from patient features.

meta

GET	/	Root	⌵
GET	/health	Health	⌵
GET	/metrics	Metrics	⌵

inference

POST	/predict	Predict	⌵
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Schemas

HTTPValidationError	⌵	Expand all	object
HealthResponse	⌵	Expand all	object
HeartFeatures	⌵	Expand all	object
PredictionResponse	⌵	Expand all	object
ValidationError	⌵	Expand all	object

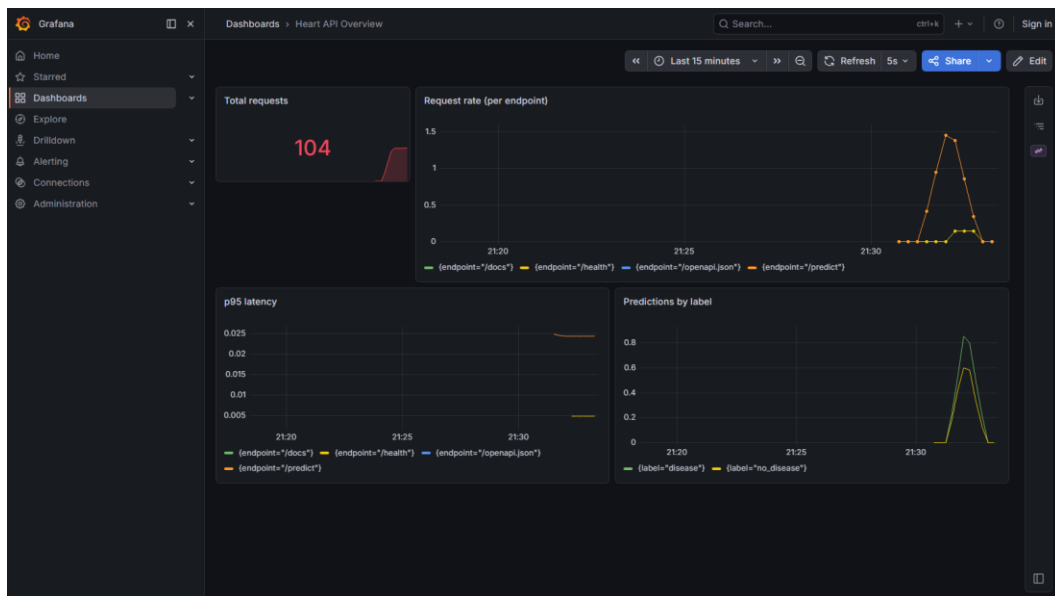
11. Monitoring and logging *(Task 8 — 3 marks)*

- **Metrics:** the API uses `prometheus_client` to expose three families on `/metrics` — `heart_api_requests_total{endpoint,method,status}`,

heart_api_request_seconds_bucket{endpoint,1e}, and

heart_api_predictions_total{label}.

- **Scraping:** monitoring/prometheus.yml declares a heart-api job scraping every 15 s.
- **Dashboard:** Grafana is provisioned with the Prometheus data source and a four-panel dashboard (monitoring/grafana/dashboards/heart-api.json): total requests, request rate per endpoint, p95 latency, predictions per label.
- **Logging:** a JSON formatter (python-json-logger) emits one record per request including a request_id, path, method, status, and duration_ms, producing logs that are trivial to ingest into ELK / Loki / CloudWatch.



Prometheus

Query

Alerts

Status > Target health

Select scrape pool

Filter by target health

Filter by endpoint or labels

heart-api

1 / 1 up

Endpoint

Labels

Last scrape

State

<http://127.0.0.1:8000/metrics>

env="local"

instance="127.0.0.1:8000"

job="heart-api"

service="heart-api"

3.864s ago

5ms

UP

prometheus

1 / 1 up

Endpoint

Labels

Last scrape

State

<http://127.0.0.1:9090/metrics>

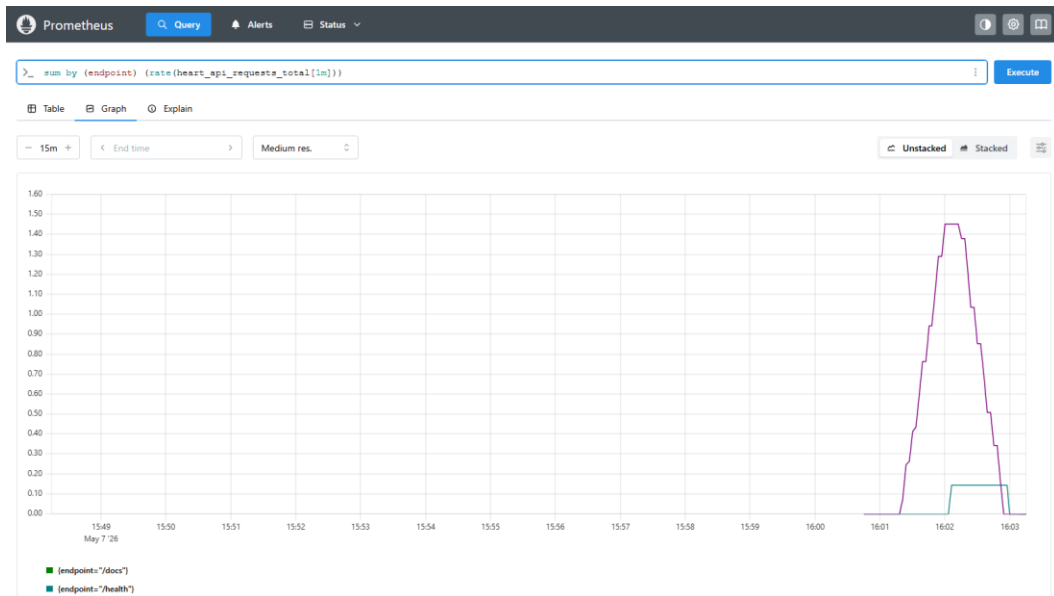
instance="127.0.0.1:9090"

job="prometheus"

3.835s ago

10ms

UP



```
# HELP python_gc_objects_collected_total Objects collected during gc
# TYPE python_gc_objects_collected_total counter
python_gc_objects_collected_total{generation="0"} 9474.0
python_gc_objects_collected_total{generation="1"} 7070.0
python_gc_objects_collected_total{generation="2"} 144.0
# HELP python_gc_objects_uncollectable_total Uncollectable objects found during GC
# TYPE python_gc_objects_uncollectable_total counter
python_gc_objects_uncollectable_total{generation="0"} 0.0
python_gc_objects_uncollectable_total{generation="1"} 0.0
python_gc_objects_uncollectable_total{generation="2"} 0.0
# HELP python_gc_collections_total Number of times this generation was collected
# TYPE python_gc_collections_total counter
python_gc_collections_total{generation="0"} 427.0
python_gc_collections_total{generation="1"} 38.0
python_gc_collections_total{generation="2"} 3.0
# HELP python_info Python platform information
# TYPE python_info gauge
python_info{implementation="CPython",major="3",minor="11",patchlevel="9",version="3.11.9"} 1.0
# HELP heart_api_requests_total API requests
# TYPE heart_api_requests_total counter
heart_api_requests_total{endpoint="/health",method="GET",status="200"} 9.0
heart_api_requests_total{endpoint="/predict",method="POST",status="200"} 93.0
heart_api_requests_total{endpoint="/docs",method="GET",status="200"} 1.0
heart_api_requests_total{endpoint="/openapi.json",method="GET",status="200"} 1.0
# HELP heart_api_requests_created API requests
# TYPE heart_api_requests_created gauge
heart_api_requests_created{endpoint="/health",method="GET",status="200"} 1.7781693340010343e+09
heart_api_requests_created{endpoint="/predict",method="POST",status="200"} 1.7781693805437882e+09
heart_api_requests_created{endpoint="/docs",method="GET",status="200"} 1.7781697825122144e+09
heart_api_requests_created{endpoint="/openapi.json",method="GET",status="200"} 1.778169787187724e+09
# HELP heart_api_request_seconds API request latency
# TYPE heart_api_request_seconds histogram
heart_api_request_seconds_bucket{endpoint="/health",le="0.005"} 9.0
heart_api_request_seconds_bucket{endpoint="/health",le="0.01"} 9.0
heart_api_request_seconds_bucket{endpoint="/health",le="0.025"} 9.0
heart_api_request_seconds_bucket{endpoint="/health",le="0.05"} 9.0
heart_api_request_seconds_bucket{endpoint="/health",le="0.1"} 9.0
heart_api_request_seconds_bucket{endpoint="/health",le="0.25"} 9.0
heart_api_request_seconds_bucket{endpoint="/health",le="0.5"} 9.0
heart_api_request_seconds_bucket{endpoint="/health",le="0.75"} 9.0
heart_api_request_seconds_bucket{endpoint="/health",le="1.0"} 9.0
heart_api_request_seconds_bucket{endpoint="/health",le="2.5"} 9.0
heart_api_request_seconds_bucket{endpoint="/health",le="5.0"} 9.0
heart_api_request_seconds_bucket{endpoint="/health",le="7.5"} 9.0
heart_api_request_seconds_bucket{endpoint="/health",le="10.0"} 9.0
heart_api_request_seconds_bucket{endpoint="/health",le="+Inf"} 9.0
heart_api_request_seconds_count{endpoint="/health"} 9.0
heart_api_request_seconds_sum{endpoint="/health"} 0.01890760082457066
heart_api_request_seconds_bucket{endpoint="/predict",le="0.005"} 0.0
heart_api_request_seconds_bucket{endpoint="/predict",le="0.01"} 4.0
heart_api_request_seconds_bucket{endpoint="/predict",le="0.025"} 99.0
heart_api_request_seconds_bucket{endpoint="/predict",le="0.05"} 99.0
heart_api_request_seconds_bucket{endpoint="/predict",le="0.075"} 93.0
heart_api_request_seconds_bucket{endpoint="/predict",le="0.1"} 93.0
heart_api_request_seconds_bucket{endpoint="/predict",le="0.25"} 93.0
heart_api_request_seconds_bucket{endpoint="/predict",le="0.5"} 93.0
heart_api_request_seconds_bucket{endpoint="/predict",le="0.75"} 93.0
heart_api_request_seconds_bucket{endpoint="/predict",le="1.0"} 93.0
```

12. Documentation and reporting *(Task 9 — 2 marks)*

- This document — docs/REPORT.md (also exported as REPORT.docx and REPORT.pdf).
- README.md covers setup, Docker, Kubernetes, Render, CI/CD, and layout.
- docs/architecture.md includes a Mermaid + ASCII architecture diagram.
- screenshots/ contains 15 PNGs referenced from this report and the

demo video (docs/2026-05-09_17-51-37.mp4).

13. Deliverables checklist

Everything the assignment asks for is included in this submission. The

table below maps each required deliverable to where it lives in the repo.

#	Deliverable	Where to find it
1	Project source code, Dockerfile and pinned dependencies	Root of the GitHub repository: https://github.com/ddmtech2025cs05056/heart-disease-mlops
2	Cleaned dataset and the script that downloads and preprocesses it	data/ and src/data/
3	Jupyter notebooks for exploratory data analysis and modelling	notebooks/01_eda.ipynb, notebooks/02_modeling.ipynb
4	Unit and integration tests (four files, nine tests, all passing)	tests/
5	GitHub Actions pipeline running lint, tests and the Docker build	.github/workflows/ci.yml
6	Kubernetes deployment manifests and Helm chart	deploy/k8s/, deploy/helm/heart-api/
7	Screenshots of every major step	screenshots/
8	Final report in three formats	docs/REPORT.pdf, docs/REPORT.docx, docs/REPORT.md

9	Short demo video walking through the whole pipeline	docs/2026-05-09 17-51-37.mp4 (also on Google Drive: https://drive.google.com/file/d/106ABOgXefeEAvoQdUCnteJ9P6vjYF_M1/view)
10	Deployed API URL (publicly reachable)	https://heart-disease-mlops-yhb7.onrender.com
11	Local run instructions	<code>kubectl port-forward svc/heart-api 8000:80</code> after applying the Kubernetes manifests

End of report.